Building figures from CSV files

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```
library(ggcpb)
```

1 Introduction

Figures can be created directly with or without using R. This vignette described the method to create figures **using only Excel**.

To create a figure, you need:

- a `params.csv` file
  - this is the parameter file, which specifies which figures to create and how they should be configured
  - one `params.csv` can contain the parameters for several figures
- a `data` folder with the dataset(s)
  - each dataset is a CSV file
- the `generate_plot.bat` script

Please note: The datasets must be saved as CSV, not as spreadsheet (`xlsx`). In Excel, choose `File` -> `Save as` -> `CSV UTF-8 (comma delimited)`.

1.1 How to create a figure without opening R?

The general workflow to create figures with ggcpb is as follows:

1. Find the most recent ggcpb version.
2. Copy it to the place where you want to create figures.
3. Put your datasets in the folder `data`. Open `params.csv` in Excel and add your parameters.
4. To create a figure, double clik on `generate_plot.bat`. This reads `params.csv`, builds every figure and saves the images in the folder `generated`. Each image is named after its dataset. Running the script again produces updated figures. Images with the same name are replaced. A dataset or figure that fails is skipped and listed at the end, and in `params_log.txt`. The other figures are still saved.

Please note. The `generate_plot.bat`, `generate_plot.R` and `params.csv` must all be in the same folder. Within this folder, there must be a subfolder `data`, in which each dataset must be saved as a CSV file.

2 The dataset

Each dataset is a CSV file in the `data` folder. It can have any name. The first row holds the column names. The package example has two datasets. One of the example datasets is shown here:

jaar	groep	koopkracht	werkloosheid
2021	werkenden	0.3	3.4
2021	gepensioneerden	-0.2	3.4
2021	uitkering	-0.9	3.4
2022	werkenden	-2.7	3.5
2022	gepensioneerden	-2.3	3.5
2022	uitkering	-3.2	3.5
2023	werkenden	1.5	3.6
2023	gepensioneerden	0.8	3.6
2023	uitkering	-0.3	3.6
2024	werkenden	1.2	3.7
2024	gepensioneerden	0.6	3.7
2024	uitkering	-0.5	3.7
2025	werkenden	0.9	3.7
2025	gepensioneerden	0.4	3.7
2025	uitkering	-0.6	3.7

The parameter file refers to variables by their column names. Categorical columns (e.g. `regio`, `geslacht`) keep the order of the rows in the created figure. So, the first group in the CSV-file (e.g. here `werkenden`) is the first on the axis and in the legend.

3 The parameter file

The parameter file determines which figures are created. Each figure has a `plot_type`, together with the settings required for that plot type. The parameter `data` names the dataset of the figure.

The first column holds the parameter names. Every other column describes one figure. This is the example parameter file. The first row, with the plot types, is the header.

plot_type	col	line
data	koopkracht	werkloosheid
title	Koopkrachtontwikkeling naar groep	Werkloosheid
x	jaar	jaar
y	koopkracht	werkloosheid
ylab	% mutatie	%

plot_type	col	line
fill	groep	
sec_y	werkloosheid	
sec_legend_label	werkloosheid	
sec_ylab	%	
position	dodge	
style		english
sec_type	line	
create	y	y

This table shows that there are two figures. The first figure (see figure 1) is a column chart (`plot_type` = `col`) that uses the `koopkracht` dataset (`data`) and that portrays `werkloosheid` on the second y-axis (`sec_y`). The second figure (see figure 2) is a line chart (`plot_type` = `line`) that uses the `werkloosheid` dataset (`data`) and has no secondary y-axis. A parameter value can be left blank, such as `sec_y` in the second plot. Only the second figure sets `style` to `english`, so it gets English labels and number formats.

The parameter `page` sets the page width a figure is saved at: `half` (7.5 cm, the default), `full` (15.5 cm) or `small`. If left out, the default `half` is used.

3.1 Plot types

`plot_type` names the figure type: `line`, `col`, `area`, `box`, `boxplot_extended`, `dot`, `scatter`, `hist`, `donut` or `map`. Each type is in R the `cpb_` function of the same name, such as `cpb_line()`. Every argument of that function can be a parameter. Appendix A lists them all per plot type.

3.2 Special values

Most values are plain text, a number, `TRUE` or `FALSE`. A few need a special form:

Value	Example	What happens
a list	`0;100`	`;` separates the values.
a line break	`First sentence\nsecond sentence`	`\n` starts a new line.
no upper limit	`0;20;Inf`	`Inf` and `-Inf` are numbers.
a formatter	`euro`	`x_labels` and `sec_axis_labels` take `number` (1.000), `euro` (€1.000) or `pct` (1%). Own labels such as `a;b;c` work too.
pairs	`p5:onderste 5%`	`dot_labels` takes name:value pairs.
a date	`2024-01-31`	A dataset column of dates gives a date axis. `31-1-2024` works too.

3.3 Classes

The parameter `cut` puts numbers into classes, such as `0;20;30;40;Inf`. That gives the classes “lager dan 20”, “20 - 30”, “30 - 40”.

With `style = english` it gives the classes “below 20”, “20 - 30”, “30 - 40” and “40 and above”.

The parameter `cut_labels` says how the class limits are written:

- `number` (default) writes plain numbers, such as “20”.
- `euro` adds a euro sign, such as “€20”.
- `pct` adds a percent sign, such as “20%”.

The classes colour the figure through one column. This is `value` of a map, `colour` of a line, dot or scatter plot, and `fill` of the other types. The numbers themselves stay available, for example on the y-axis.

3.4 Saving

Next to `page`, four parameters say how a figure is saved:

Parameter	Example	What happens
`height`	`10`	The height in cm. Left out, the page height is used.
`preset`	`map`	`map` makes a map as tall as it needs. `presentation` is for slides.
`dpi`	`800`	The resolution. The default is 300.
`lock`	`FALSE`	A title that is too wide only gives a warning, instead of an error.

4 Creating the figures using R

Pass the parameter file to `cpb_import_csv()`. The datasets are read from the `data` folder next to it:

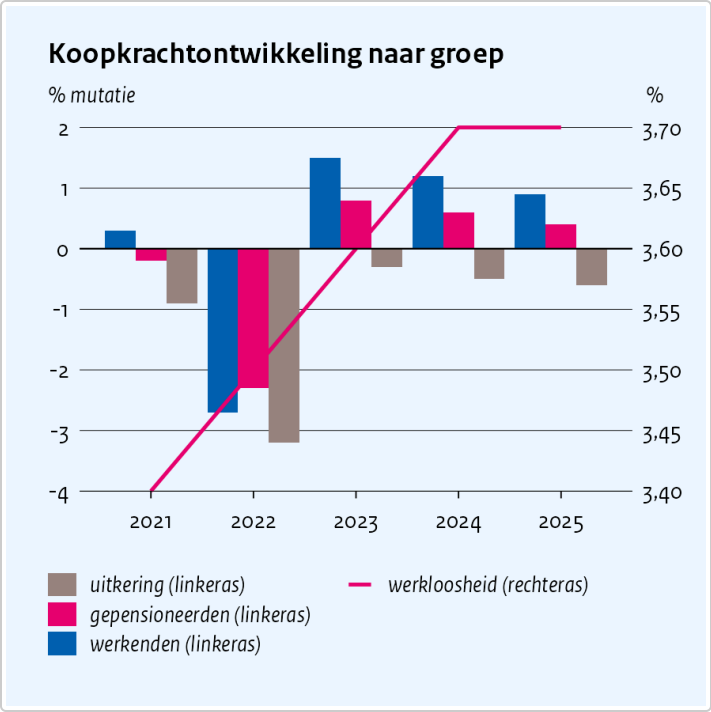
```
figure_dir <- file.path(tempdir(), "figures")
params_csv <- file.path(figure_dir, "params.csv")
figs <- cpb_import_csv(params_csv)
```

When the parameter file contains multiple figures, `cpb_import_csv()` returns a named list. The names come from the datasets. Here the two figures use different datasets, so they are called `koopkracht` and `werkloosheid`.

`save_cpb()` precisely positions the secondary-axis caption in the `koopkracht` figure, so the figure is saved using `save_cpb()` rather than printed directly.

```
path <- tempfile(fileext = ".png")
save_cpb(path, figs$koopkracht, page = "half")
```

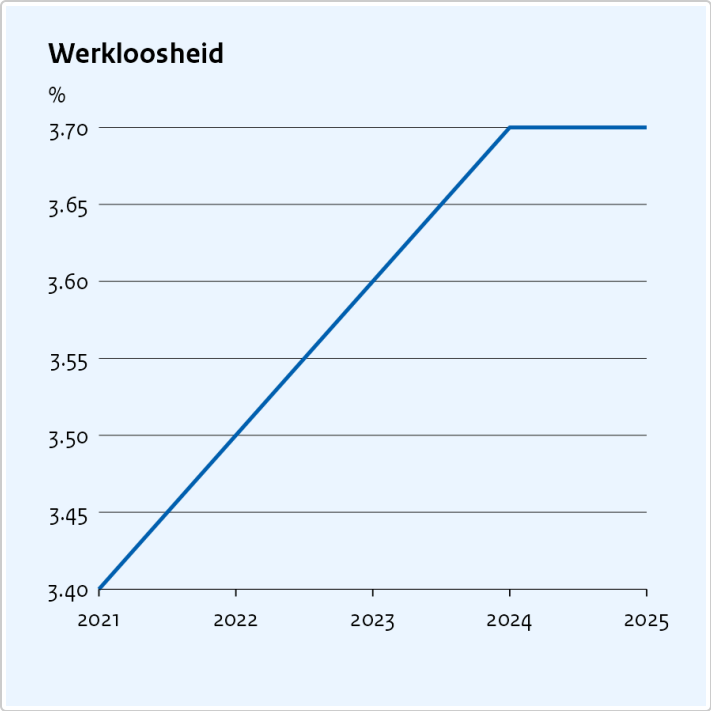
Figure 1. The column chart of the first figure in params.csv



`werkloosheid` has no second axis, so it can be shown directly:

Figure 2. The line chart of the second figure in params.csv

```
figs$werkloosheid
```



5 Several datasets

The parameter `data` names the dataset of a figure, relative to the `data` folder next to `params.csv`. The extension `.csv` is optional, so `koopkracht` and `koopkracht.csv` are the same dataset.

Parameter *data*	What happens
`koopkracht`	The figure uses `data/koopkracht.csv`.
`koopkracht;werkloosheid`	Two figures are made: one from `data/koopkracht.csv` and one from `data/werkloosheid.csv`.
blank	The figure uses `data/data.csv`.
column absent	The same as blank.

The figures are named after their dataset. If several figures use the same dataset, the figure number is added to the name. The first figure column in `params.csv` is number 1. For example, `koopkracht_1.png` and `koopkracht_3.png`.

A named dataset that does not exist in the `data` folder is skipped with a warning. Nothing falls back to another dataset. The other datasets in the same list and the other rows are still created. With `data` set to `a;b;c` and no `b.csv`, you get figures for `a` and `c`, and a warning about `b.csv`.

5.1 A parameter file with one figure

A parameter file can describe a single figure or several figures. When describing only one figure, it has one column of values next to the parameter names. For example, the following describes a line chart. It can be saved in the same folder, so it finds the same `data` folder:

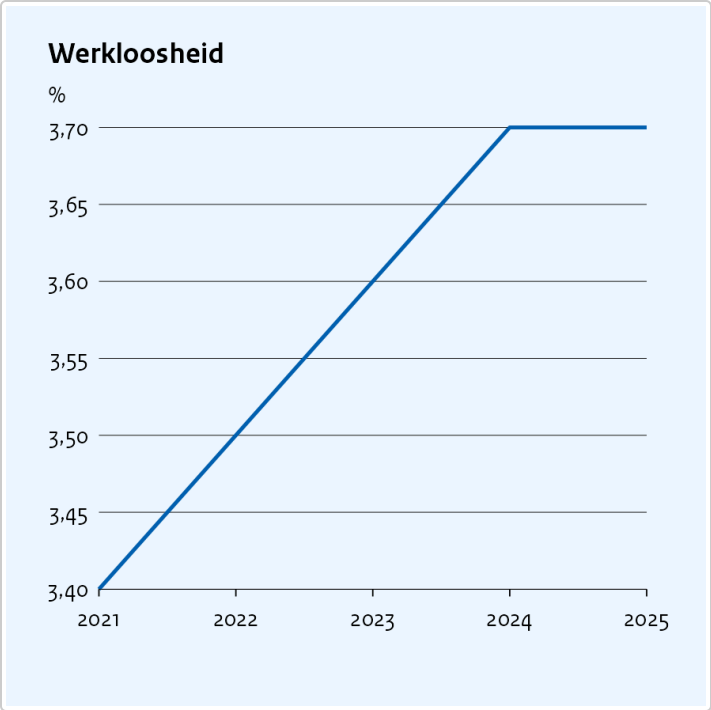
```
params_one <- file.path(figure_dir, "params_one.csv")

writeLines(c(
  "plot_type,line",
  "title,Werkloosheid",
  "data,werkloosheid",
  "x,jaar",
  "y,werkloosheid",
  "ylab,%"
), params_one)

cat(readLines(params_one), sep = "\n")
#> plot_type,line
#> title,Werkloosheid
#> data,werkloosheid
#> x,jaar
#> y,werkloosheid
#> ylab,%
```

When the parameter file describes one figure, `cpb_import_csv()` returns that figure directly rather than a named list. Use `simplify = FALSE` to always get a named list.

```
fig_one <- cpb_import_csv(params_one)
```



6 Handling problems

A parameter can be left blank. In that case, the default value is used.

If a parameter is not valid for the selected `plot_type`, it is ignored and the default is used. This produces a warning.

If a figure cannot be created, for example because a column referenced by a parameter does not exist in the dataset, that figure is skipped. Other figures in the same parameter file are still created.

Each call to `cpb_import_csv()` also writes a log file next to the parameter file. The log records parameter problems and figures that were created or skipped.

Appendix A: Parameters

A parameter that refers to a variable (such as `x`, `y` or `fill`) takes a column name from the dataset. Every other parameter takes a value directly from the parameter file.

The available parameters depend on the `plot_type`. Table A1 lists every parameter once, with the plot types it belongs to, its default, an example and a short description. A default that differs per plot type is followed by the plot types it holds for.

Table A1. All parameters

Parameter	Plot types	Default	Example	Description
axis_text_size	col, line, box, dot, scatter, hist, boxplot_extended	7	7	The size of the axis text, in points.
bins	hist	none	30	The number of bins.
binwidth	hist	none	5	The width of one bin of the histogram.
border_colour	map	none	white	The colour of the region borders.

Parameter	Plot types	Default	Example	Description
border_linewidth	map	0.15	0.15	The width of the region borders.
box_labels	box, boxplot_extended	none	TRUE	TRUE prints the values of the box.
box_style	box, boxplot_extended	modern: boxplot_extended ggcpb: rest	`ggcpb`, `james`, `modern` or `dot`	The look of the boxes.
cap_width	dot	0.25	0.25	The width of the end caps of the interval.
colour	line, dot, scatter	none	group	The column that gives lines or points their colour.
colour_index	all	none	2;6	Which house colours to use, by number. The first number is for the first category.
cut	all	none	0;20;40;Inf	Cuts the numbers of the colour column into classes. The values are the class limits.
cut_labels	all	number	`number`, `euro` or `pct`	How the class limits are written in the legend.
data	all	data	koopkracht	Excel workflow: CSV file in the data folder. Leave blank for data.csv. Write a;b for one figure per dataset.
dot_labels	box, boxplot_extended	none	p5:bottom 5%;p95:top 5%	Names for the dots of the dot style, as name:text pairs.
dpi	all	300	800	The sharpness of the saved image, in dots per inch.
facet	col, line, box, dot, scatter, hist, boxplot_extended	none	region	The column that splits the figure into panels. The number of panels equals the unique values of that column.
facet_ncol	col, line, box, dot, scatter, hist, boxplot_extended	none	2	The maximum number of panels next to each other per row.
facet_scales	col, line, box, dot, scatter, hist, boxplot_extended	fixed	`fixed`, `free`, `free_x` or `free_y`	Whether the panels share their axes (fixed) or each have their own (free).
fill	col, box, hist, donut, boxplot_extended	required	group	The column that splits the bars or areas into colours.
fill_colour	col, box, hist, boxplot_extended	none	#005faf	One colour for all bars or areas. It is used when `fill` is not set.
flush_legend	all	TRUE	TRUE	TRUE places the legend at the left edge of the figure.
forecast_label	col, line, scatter	raming	raming	The word shown in the forecast area.

Parameter	Plot types	Default	Example	Description
forecast_x	col, line, scatter	none	2025	The x value where the forecast starts. Everything after it is shaded.
grid_colour	col, line, box, dot, scatter, hist, boxplot_extended	white: boxplot_extended black: rest	black	The colour of the grid lines.
grid_linewidth	col, line, box, dot, scatter, hist, boxplot_extended	0.8: boxplot_extended 0.1: rest	0.1	The width of the grid lines.
group	col, box, dot, boxplot_extended	none	sector	The column that splits the data into groups.
group_gap	col, box, dot, boxplot_extended	0.8: col 0.7: rest	0.8	The space between the groups, in bar widths.
height	all	none	10	The height of the saved figure in cm. Blank gives the standard height.
label	donut	none	name	The column with the text of each slice.
label_accuracy	box, donut, boxplot_extended	1: donut 0.1: rest	0.1	The rounding of the printed percentages. 0.01 allows for two decimals.
label_colour	donut	black	black	The colour of the labels in the slices.
label_style	donut	wedge	`wedge` or `leader`	Prints the label inside the slice (wedge) or outside with a line (leader).
leader_length	donut	0.15	0.15	The length of the lines to the labels.
legend	all	topleft: map bottom: rest	`bottom`, `right`, `left`, `top` or `none`	Where the legend is placed. none removes it.
legend_ncol	all	none	2	The number of columns of the legend.
legend_nrow	all	none	2	The number of rows of the legend.
legend_pct	donut	FALSE	FALSE	TRUE adds the percentage to each name in the legend.
legend_title	all	none	Legend	The title of the legend.
level	map	gemeente	`gemeente`, `corop` or `provincie`	The kind of regions on the map.
line_colour	line	none	#005faf	One colour for the whole line. It is used when `colour` is not set.

Parameter	Plot types	Default	Example	Description
linewidth	line, box, dot, boxplot_extended	0.55: line 0.4: dot 0.25: rest	0.55	The width of the line.
lock	all	TRUE	FALSE	TRUE stops with a message when the title or legend is wider than the page. FALSE saves the figure anyway.
lower	dot	required	lower_bound	The column with the lower bound of the interval.
mean	box, boxplot_extended	none	mean	The column with the mean, drawn as a diamond.
minor	col, line, box, dot, scatter, hist, boxplot_extended	FALSE	FALSE	TRUE adds small ticks between the labels.
na_fill	map	none	lightgrey	The colour of regions without a value.
orientation	col, box, dot, boxplot_extended	vertical: col, box horizontal: rest	`vertical` or `horizontal`	Draws the figure upright (vertical) or turned (horizontal).
outline	hist	white	white	The colour of the line around the bars.
p25	box, boxplot_extended	required	p25	The column with the 25th percentile, the bottom of the box.
p5	box, boxplot_extended	required	p5	The column with the 5th percentile, the lower whisker.
p50	box, boxplot_extended	required	p50	The column with the median, the line in the box.
p75	box, boxplot_extended	required	p75	The column with the 75th percentile, the top of the box.
p95	box, boxplot_extended	required	p95	The column with the 95th percentile, the upper whisker.
page	all	half	`half`, `full` or `small`	The width of the saved figure: half is 7.5 cm, full is 15.5 cm and small is 6.8 cm.
palette	all	sequential: map qualitative: rest	`qualitative`, `discr`, `sequential`, `blues` or `extended`	The colour set used for the categories.
panel_fill	boxplot_extended	#eef8ff	#eef8ff	The colour behind the figure.
panel_size	donut	none	1.8	The size of the figure area in inches. It keeps the ring the same size.

Parameter	Plot types	Default	Example	Description
pct_axis	col, line, box, dot, scatter, boxplot_extended	FALSE	FALSE	TRUE writes the values on the axis as percentages.
points	line	FALSE	FALSE	TRUE adds points to the line.
position	col, hist	stack	`stack`, `dodge` or `fill`	How bars of one category are placed: on top of each other (`stack`), next to each other (`dodge`) or as shares of 100% (`fill`).
preset	all	report	`report`, `presentation` or `map`	The height of the figure. `report` is 7.5 cm (6.8 cm for `small`), `presentation` is 6.35 cm and `map` is as tall as the map needs.
region	map	required	code	The column with the codes of the municipalities, COROP regions or provinces.
reverse	map	FALSE	FALSE	TRUE reverses the colours of the gradient of a numeric `value` column.
reverse_legend	all	TRUE: col, hist FALSE: rest	TRUE	TRUE reverses the legend order.
ring_width	donut	0.6	0.6	The thickness of the ring, as a share of the radius.
sec_accuracy	col, line, box, dot, boxplot_extended	none	0.1	The rounding of the numbers on the secondary axis. 0.001 gives three decimals.
sec_at	col, line, box, dot, boxplot_extended	none	0;50;100	The values at which the secondary axis is labelled.
sec_axis_labels	col, line, box, dot, boxplot_extended	none	`pct`, `number`, `euro` or labels such as `a;b;c`	How the numbers on the secondary axis are written, or your own labels. With a;b;c the ticks get the labels a, b and c, so use as many labels as ticks (see sec_at).
sec_col_width	col, line, box, dot, boxplot_extended	0.3	0.3	The width of the bars of the second series.
sec_colour	col, line, box, dot, boxplot_extended	none	#e6006e	The colour of the second series.
sec_legend_label	col, line, box, dot, boxplot_extended	none	name of the second series	The name of the second series in the legend.
sec_limits	col, line, box, dot, boxplot_extended	none	0;100	The lowest and highest value of the secondary axis.
sec_linewidth	col, line, box, dot, boxplot_extended	none: line 0.55: rest	0.55	The line width of the second series.

Parameter	Plot types	Default	Example	Description
sec_points	col, line, box, dot, boxplot_extended	FALSE	FALSE	TRUE adds points to the second series when it is a line.
sec_scale_auto	col, line, box, dot, boxplot_extended	TRUE	TRUE	TRUE chooses the secondary axis automatically, so its ticks line up with the gridlines of the value axis.
sec_symbol_size	col, line, box, dot, boxplot_extended	symbol_size: dot 1.6: rest	1.6	The size of the points of the second series.
sec_type	col, line, box, dot, boxplot_extended	line	`line`, `point` or `col`	How the second series is drawn.
sec_y	col, line, box, dot, boxplot_extended	none	unemployment	The column drawn on a secondary axis.
sec_ylab	col, line, box, dot, boxplot_extended	none	unit of the secondary axis	The unit above the secondary axis.
style	all	dutch	`dutch` or `english`	The language of the figure: dutch or english. It sets the number format (1.000,5 or 1,000.5), the forecast label (raming or forecast), the axis names in the legend (linkeras and rechteras, or left axis and right axis) and the class labels of cut.
subtitle	all	none	subtitle (replaces the default unit above the figure)	The text under the title. It replaces the unit.
symbol_colour	dot, scatter	none	#e6006e	The colour of the dots.
symbol_size	line, dot, scatter	1.1: line 1.4: dot 0.8: rest	1.1	The size of the points.
ticks	col, line, box, dot, scatter, hist, boxplot_extended	FALSE: boxplot_extended TRUE: rest	TRUE	TRUE draws ticks on the axis.
title	all	none	Title of the figure	The title above the figure.
upper	dot	required	upper_bound	The column with the upper bound of the interval.
value	map	required	share	The column with the values that give the regions their colour.
value_accuracy	col, line, box, dot, scatter, hist, boxplot_extended	none	0.1	The rounding of the numbers on the value axis, such as 0.1.

Parameter	Plot types	Default	Example	Description
value_axis	box, boxplot_extended	none: boxplot_extended bottom: rest	`bottom` or `top`	Whether the value axis is at the bottom or at the top.
value_axis_linewidth	boxplot_extended	0.7	0.7	The width of the line of the value axis.
value_breaks	col, line, box, dot, scatter, hist, boxplot_extended	none	0;25;50;75;100	The values at which the value axis is labelled.
value_crop	col, line, box, dot, scatter, hist, boxplot_extended	FALSE	FALSE	TRUE cuts off the data outside the value limits, with a warning.
value_labels	col	FALSE	FALSE	TRUE prints the value on each bar.
value_limits	col, line, box, dot, scatter, hist, boxplot_extended	none	0;100	The lowest and highest value of the value axis. Note: if value_crop is not TRUE, the axis widens when the data runs past these limits.
wedge_labels	donut	TRUE	TRUE	TRUE prints the value on each slice.
width	box, boxplot_extended	0.5: box 0.45: rest	0.5	The width of the bars or boxes, as a share of the space.
x	col, line, box, dot, scatter, hist, boxplot_extended	required	year	The column with the categories or numbers on the x-axis.
x_labels	scatter	none	`euro`, `number`, `pct` or labels such as `a;b;c`	How the numbers on the x-axis are written, or your own labels. With a;b;c the ticks get the labels a, b and c, so use as many labels as ticks.
x_lim	col, line, box, dot, scatter, hist, boxplot_extended	none	2015;2025	The first and last value shown on the x-axis.
x_lim_follow_data	col, line, box, dot, scatter, hist, boxplot_extended	FALSE: col, scatter, hist TRUE: rest	FALSE	TRUE lets the x-axis follow the data when it runs past x_lim.
x_ticks_ends	col, line	FALSE	FALSE	TRUE also labels the first and last month of a monthly axis.
xlab	col, line, box, dot, scatter, hist, boxplot_extended	none	unit below the x-axis	The text next to the x-axis.
y	col, line, dot, scatter, donut	required	purchasing_power	The column with the values.
ylab	col, line, box, dot, scatter, hist, boxplot_extended	none	% change	The unit above the value axis.

Parameter	Plot types	Default	Example	Description
ylab_position	boxplot_extended	none	`left`, `middle` or `right`	Where the unit is placed: left, middle or right.
ymax	line	none	upper_bound	The column with the upper end of an uncertainty band.
ymin	line	none	lower_bound	The column with the lower end of an uncertainty band.
zero_indicator	boxplot_extended	none	TRUE	TRUE draws a thick white line at zero behind the data.
zero_indicator_linewidth	boxplot_extended	2	2	The width of the white zero line.
zeroline	col, line, box, dot, scatter, hist	TRUE: col, hist none: rest	TRUE	TRUE draws a line at zero. Blank draws it only when zero is on the axis.